

Chapter 13

Microscopic Model for Ideal gas

13.1 Ideal gas

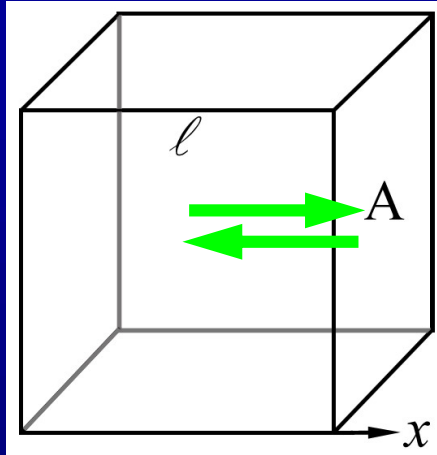
Macroscopic description $pV = nRT$

$$C_V = \frac{3}{2}nR \quad \dots$$

Microscopic description

- It consists of N identical molecules, N is large.
- The molecules obey Newton's law.
- The average spacing $\gg r$
- Collisions are elastic and are of negligible duration.

microscopic meaning of **pressure**



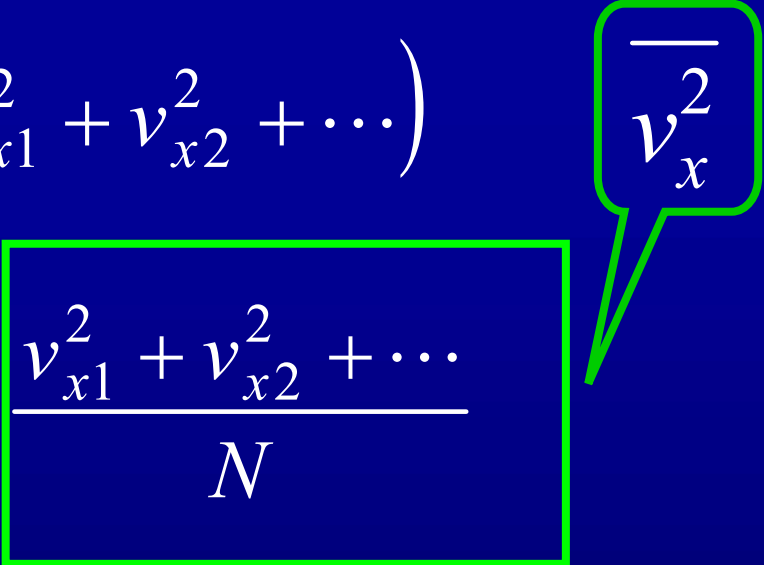
$$\begin{aligned}\Delta p_x &= p_f - p_i = (-mv_x) - mv_x \\ &= -2mv_x\end{aligned}$$

$$\Delta t = \frac{2\ell}{v_x}$$

$$\overline{F} = -\frac{\Delta p_x}{\Delta t} \text{ (the force on wall A)}$$

$$= \frac{2mv_x}{\frac{2\ell}{v_x}} = \frac{mv_x^2}{\ell}$$

$$\frac{1}{\ell^2} \sum_{n=1}^N F_i = \frac{m}{\ell^3} (v_{x1}^2 + v_{x2}^2 + \dots)$$

$$= \frac{Nm}{\ell^3} \times \frac{v_{x1}^2 + v_{x2}^2 + \dots}{N}$$


$$\frac{1}{\ell^2} \sum_{n=1}^N F_i = \frac{Nm}{V} \times \frac{n_1 v_{x(1)}^2 + n_2 v_{x(2)}^2 + \dots}{n_1 + n_2 + \dots}$$

$$p = \frac{Nm}{V} \overline{v_x^2} = \frac{1}{3} \frac{Nm}{V} \overline{v^2} = \frac{1}{3} \rho \overline{v^2}$$

$$v_{\text{rms}} \equiv \sqrt{v^2} = \sqrt{\frac{3p}{\rho}}$$

$$v_{\text{s}} = \sqrt{\frac{\gamma p}{\rho}} = \sqrt{\frac{\gamma}{3}} v_{\text{rms}} \approx 0.68 v_{\text{rms}}$$

Example 13.1 Sound speed in air

$$\rho = \frac{Nm}{V} = \frac{nN_A m}{V} = \frac{nM}{V} = \frac{M}{V_{m0}}$$

or
$$\bar{M} = \left(28 \times \frac{4}{5} + 32 \times \frac{1}{5} \right) \text{kg/kmol} = 28.8 \text{kg/kmol}$$

$$\begin{aligned} \bar{M} &= (28 \times 76.9\% + 32 \times 23.1\%) \text{kg/kmol} \\ &= 28.9 \text{kg/kmol} \end{aligned}$$

$$V_{m0} = 22.41383 \times 10^{-3} \text{ m}^3/\text{mol}$$

(*at* 273.15K, $1.0135 \times 10^5 \text{ Pa}$)

$$v_{\text{rms}} = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3 \times 1.0135 \times 10^5}{\frac{28.9/10^3}{22.41383 \times 10^{-3}}}} \approx 483 \text{ m/s}$$

$$v_s = \sqrt{\frac{1}{3} \times \frac{7}{5}} v_{\text{rms}} \approx 331 \text{ m/s}$$

microscopic **interpretation** of T and U

$$pV = \frac{2}{3}N\left(\frac{1}{2}m\overline{v^2}\right) \qquad p = \frac{1}{3}\frac{Nm}{V}\overline{v^2}$$

According to equation of state

$$nRT = \frac{2}{3}N\left(\frac{1}{2}m\overline{v^2}\right)$$

$$\frac{1}{2}m\overline{v^2} = \frac{3}{2}\frac{nR}{N}T = \frac{3}{2}\frac{R}{N_A}T = \frac{3}{2}k_B T \quad \text{classical}$$

$$U = N \left(\frac{1}{2} m \overline{v^2} \right) + U_0 = \frac{3}{2} nRT + U_0$$

$$C_V = \left(\frac{dU}{dT} \right)_V = \frac{3}{2} nR$$

For diatomic gas

$$\frac{U}{n} = \frac{3}{2} RT + 2 \times \frac{1}{2} RT + \frac{U_0}{n}$$

Two rotational degree of freedom

13.2 Equilibrium distribution

$$\overline{v^2} = \frac{v_1^2 + v_2^2 + \dots + v_N^2}{N} = \frac{n_1 v_1'^2 + n_2 v_2'^2 + \dots}{N}$$

$$= \sum_i v_i'^2 \left(\frac{n_i}{N} \right) \rightarrow \int v^2 \frac{dn}{N}$$

n_i — the number of molecules of v_i'

dn is the number of molecules in $v_x \rightarrow v_x + dv_x$

near v

$$v_y \rightarrow v_y + dv_y$$

$$v_z \rightarrow v_z + dv_z$$

Meaning of dn/N

among N molecules, how many near $\mathbf{v}$

It is the probability that **one molecule** near $\mathbf{v}$

proportional to $d\mathbf{v}_x \cdots$

$$f(v_x)dv_x$$

$$\frac{dn(v_x, v_y, v_z)}{N} = f(v_x)dv_x \cdot f(v_y)dv_y \cdot f(v_z)dv_z$$

Isotropic assumption

$$f(v_x)f(v_y)f(v_z) = \varphi(v^2) = \varphi(v_x^2 + v_y^2 + v_z^2)$$

The simplest solution is

$$f(v_x) = C \exp(-v_x^2 / \alpha^2)$$

The probability satisfies normalization

$$1 = \int \frac{dn}{N}$$

$$1 = \int \frac{dn}{N} = C^3 \left[\int \exp(-v_x^2 / \alpha^2) dv_x \right]^3$$

$$1 = C \int \exp(-v_x^2 / \alpha^2) dv_x$$

$$C = \frac{1}{\int_{-\infty}^{\infty} \exp\left(-\frac{v_x^2}{\alpha^2}\right) \frac{dv_x}{\alpha}} = \frac{1}{\alpha \sqrt{\pi}}$$

Gaussian integral $I = \int_0^{\infty} e^{-x^2} dx$

$$\left[\int_{-\infty}^{\infty} e^{-x^2} dx \right]^2 = \iint e^{-(x^2+y^2)} dx dy = \int_0^{\infty} e^{-r^2} 2\pi r dr = \pi$$

$$I = \frac{\sqrt{\pi}}{2}$$

Using gamma function $\Gamma(\nu) = \int_0^{\infty} e^{-t} t^{\nu-1} dt$

$$I = \int_0^{\infty} e^{-y} \frac{1}{2\sqrt{y}} dy = \frac{1}{2} \Gamma\left(\frac{1}{2}\right), \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

For $\nu > 0$,

$$\Gamma(\nu + 1) = \int_0^{\infty} e^{-t} t^{\nu} dt = - \int_0^{\infty} t^{\nu} d(e^{-t})$$

$$= - t^{\nu} e^{-t} \Big|_0^{\infty} + \int_0^{\infty} e^{-t} d(t^{\nu})$$

$$= \nu \int_0^{\infty} e^{-t} t^{\nu-1} dt = \nu \Gamma(\nu).$$

$$\Gamma(1) = \int_0^{\infty} e^{-t} t^{1-1} dt = \int_0^{\infty} e^{-t} dt = - e^{-t} \Big|_0^{\infty} = 1$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

For $n \in N^*$, $\Gamma(n+1) = n\Gamma(n) = n!$;

$$\begin{aligned}\Gamma\left(n + \frac{1}{2}\right) &= \left(n - \frac{1}{2}\right) \cdots \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{(2n-1)(2n-3) \cdots 1}{2^n} \sqrt{\pi} \\ &= \frac{(2n-1)!!}{2^n} \sqrt{\pi}.\end{aligned}$$

$$I_n = \int_0^{\infty} x^n e^{-x^2} dx = \frac{1}{2} \int_0^{\infty} e^{-y} y^{\frac{n-1}{2}} dy = \frac{1}{2} \Gamma\left(\frac{n+1}{2}\right)$$

$$I_{2m-1} = \frac{1}{2} \Gamma(m) = \frac{1}{2} (m-1)!$$

$$I_{2m} = \frac{1}{2} \Gamma\left(m + \frac{1}{2}\right) = \frac{(2m-1)!!}{2^{m+1}} \sqrt{\pi}$$

$$\begin{aligned}
I_{2m} &= \int_0^{\infty} x^{2m} e^{-x^2} dx \\
&= \left[(-1)^m \frac{\partial^m}{\partial \alpha^m} \int_0^{\infty} e^{-\alpha x^2} dx \right]_{\alpha=1} \\
&= \frac{1}{2} \left[(-1)^m \frac{\partial^m}{\partial \alpha^m} \sqrt{\frac{\pi}{\alpha}} \right]_{\alpha=1} \\
&= \frac{1}{2} \frac{1}{2} \frac{3}{2} \cdots \frac{2m-1}{2} \sqrt{\pi}
\end{aligned}$$

$$\frac{dn(v_x, v_y, v_z)}{N}$$

$$= \left(\frac{1}{\alpha \sqrt{\pi}} \right)^3 \exp \left[- \frac{(v_x^2 + v_y^2 + v_z^2)}{\alpha^2} \right] dv_x dv_y dv_z$$

$$\alpha = ?$$

Known $\overline{v^2} = \frac{3k_B T}{m}$

while $\overline{v^2} = \int v^2 \frac{dn(v_x, v_y, v_z)}{N} = \frac{3}{2} \alpha^2$

$$\frac{dn(v_x, v_y, v_z)}{N}$$

$$= \left(\frac{1}{\alpha\sqrt{\pi}} \right) e^{-\frac{v_x^2}{\alpha^2}} dv_x \left(\frac{1}{\alpha\sqrt{\pi}} \right) e^{-\frac{v_y^2}{\alpha^2}} dv_y \left(\frac{1}{\alpha\sqrt{\pi}} \right) e^{-\frac{v_z^2}{\alpha^2}} dv_z$$

$$\overline{v^2} = \int \left(v_x^2 + v_y^2 + v_z^2 \right) \frac{dn(v_x, v_y, v_z)}{N}$$

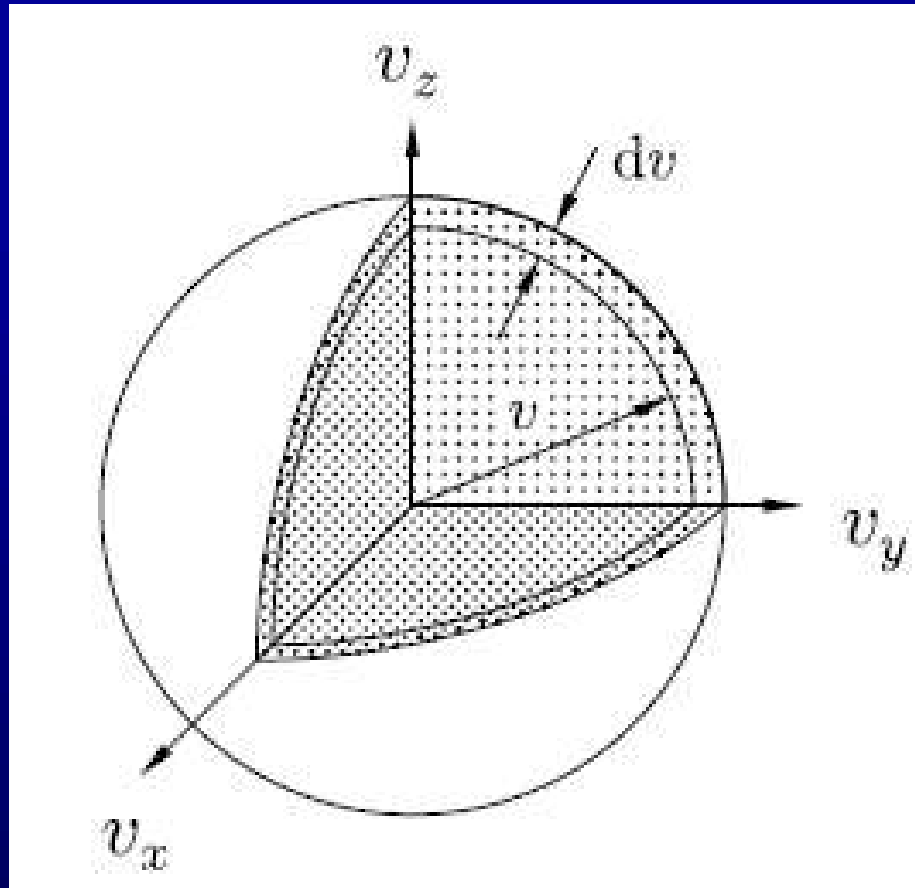
$$= 3 \int_{-\infty}^{\infty} v_x^2 \left(\frac{1}{\alpha\sqrt{\pi}} \right) e^{-\frac{v_x^2}{\alpha^2}} dv_x = 3 \times \frac{2\alpha^2}{\sqrt{\pi}} \int_0^{\infty} y^2 e^{-y^2} dy$$

$$\frac{dn(v_x, v_y, v_z)}{N} = \left(\frac{m}{2\pi k_B T} \right)^{3/2} \exp \left\{ -\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T} \right\} dv_x dv_y dv_z$$

For unit volume

$$dn(v_x, v_y, v_z) = n_0 \left(\frac{m}{2\pi k_B T} \right)^{3/2} \exp \left\{ -\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T} \right\} dv_x dv_y dv_z$$

Maxwell velocity distribution



$$dv_x dv_y dv_z \rightarrow 4\pi v^2 dv$$

$$dn(v) = n_0 \left(\frac{m}{2\pi k_B T} \right)^{3/2} \exp \left\{ -\frac{\frac{1}{2}mv^2}{k_B T} \right\} 4\pi v^2 dv$$
$$\equiv f(v)dv$$

Maxwell speed distribution

Most probable speed

For **one molecule**, what speed is most possible?

Among **N molecules**, what is the speed most molecules have in unit interval?

$$\frac{d}{dv} f(v) = 0 = \frac{d}{dv} \left[v^2 \exp \left(-\frac{1}{2} \frac{mv^2}{k_B T} \right) \right]$$

$$v_m = \sqrt{\frac{2k_B T}{m}}$$

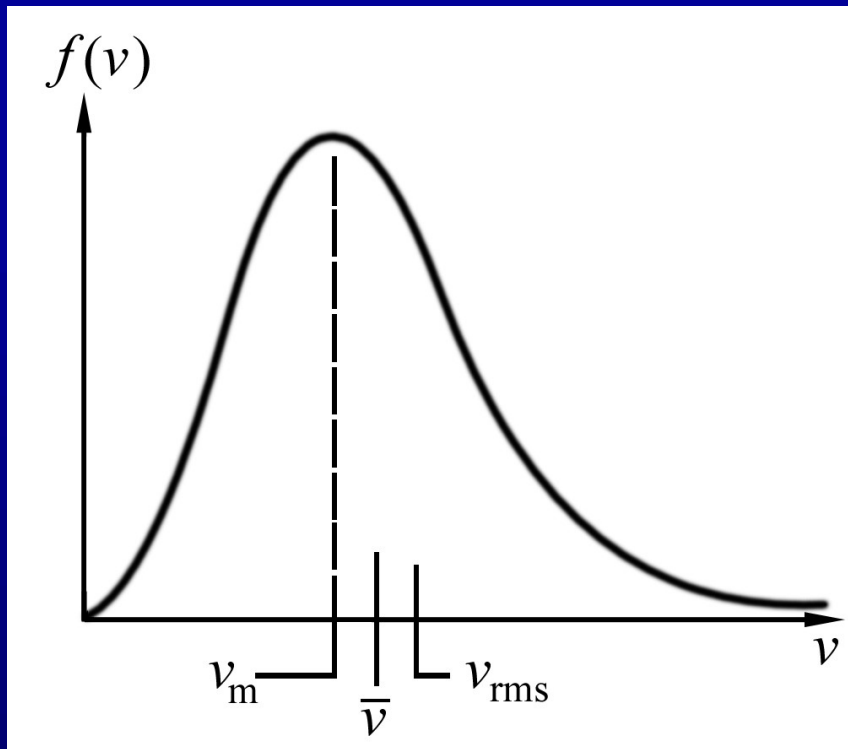
Maxwellian can be written as

$$\frac{dn(v)}{N} = 4\pi \left(\frac{1}{\pi v_m^2} \right)^{3/2} \exp \left(-\frac{v^2}{v_m^2} \right) v^2 dv$$

The average speed

$$\begin{aligned}\bar{v} &= 4\pi \left(\frac{1}{\pi v_m^2} \right)^{3/2} \int_0^{\infty} v \exp(-v^2/v_m^2) v^2 dv \\ &= \frac{4}{\sqrt{\pi}} v_m \int_0^{\infty} x^3 e^{-x^2} dx = \frac{2}{\sqrt{\pi}} \sqrt{\frac{2k_B T}{m}} \Gamma(2) \\ &= \sqrt{\frac{8k_B T}{\pi m}}\end{aligned}$$

$$\begin{aligned}
 \bar{v} &= \frac{\cancel{4\pi} \left(\cancel{\frac{1}{\pi v_m^2}} \right)^{3/2} \int_0^\infty v \exp(-v^2/v_m^2) v^2 dv}{\cancel{4\pi} \left(\cancel{\frac{1}{\pi v_m^2}} \right)^{3/2} \int_0^\infty \exp(-v^2/v_m^2) v^2 dv} \\
 &= v_m \frac{\int_0^\infty x \exp(-x) \cdot \cancel{\frac{1}{2}} dx}{\int_0^\infty x^{\frac{1}{2}} \exp(-x) \cancel{\frac{1}{2}} dx} = v_m \frac{\Gamma(2)}{\Gamma(3/2)}
 \end{aligned}$$



$$\left\{ \begin{array}{l} v_m = \sqrt{\frac{2k_B T}{m}} = 1.41 \sqrt{\frac{k_B T}{m}} \\ \bar{v} = \sqrt{\frac{8k_B T}{\pi m}} = 1.59 \sqrt{\frac{k_B T}{m}} \\ v_{\text{rms}} = \sqrt{\frac{3k_B T}{m}} = 1.73 \sqrt{\frac{k_B T}{m}} \end{array} \right.$$

$$\overline{v^\alpha} = \frac{1}{n_0} \int v^\alpha f(v) dv = \frac{\int v^\alpha f(v) dv}{\int f(v) dv}$$

Assignment: 13.2, 13.6

In problem 13.6

$$\frac{2}{\sqrt{\pi}} \int_0^1 e^{-x^2} dx = \operatorname{erf}(1) = 0.84270$$

13.3 Equipartition theorem

Maxwellian

$$\frac{dn(v)}{n_0} = \left(\frac{m}{2\pi k_B T} \right)^{3/2} \exp \left\{ -\frac{\frac{1}{2}mv^2}{k_B T} \right\} 4\pi v^2 dv$$

is a special example of **Boltzmann distribution**:

$$\frac{e^{-\beta E(p,q)} dp dq}{\iint e^{-\beta E(p,q)} dp dq} \quad \left(\beta = \frac{1}{k_B T} \right)$$

where the energy can be

$$E = \frac{1}{2m} p^2 + mgz \quad \text{translation + gravity}$$

$$E = \frac{1}{2m} p^2 + \frac{1}{2} m \omega^2 x^2 \quad \text{SHO}$$

$$E = Ap^2 + Bq^2$$

$$\overline{Ap^2} = \frac{\iint Ap^2 e^{-\beta(Ap^2 + Bq^2)} dp dq}{\iint e^{-\beta(Ap^2 + Bq^2)} dp dq}$$

$$= A \frac{\left[-\frac{\partial}{\partial(\beta A)} \right] \left[\iint e^{-\beta(Ap^2 + Bq^2)} dp dq \right]}{\iint e^{-\beta(Ap^2 + Bq^2)} dp dq}$$

$$= A \left[-\frac{\partial}{\partial(\underline{\beta A})} \right] \ln \left[\int e^{-\underline{\beta A} p^2} dp \int e^{-\beta B q^2} dq \right]$$

$$= A \left[-\frac{\partial}{\partial(\beta A)} \right] \ln \left[\frac{1}{\sqrt{\beta A}} \int e^{-\beta A p^2} d(\sqrt{\beta A} p) \right]$$

$$= A \left[-\frac{\partial}{\partial(\beta A)} \right] \left(-\frac{1}{2} \ln \beta A \right)$$

$$= \frac{1}{2\beta} = \frac{1}{2} k_{\text{B}} T$$

Similarly

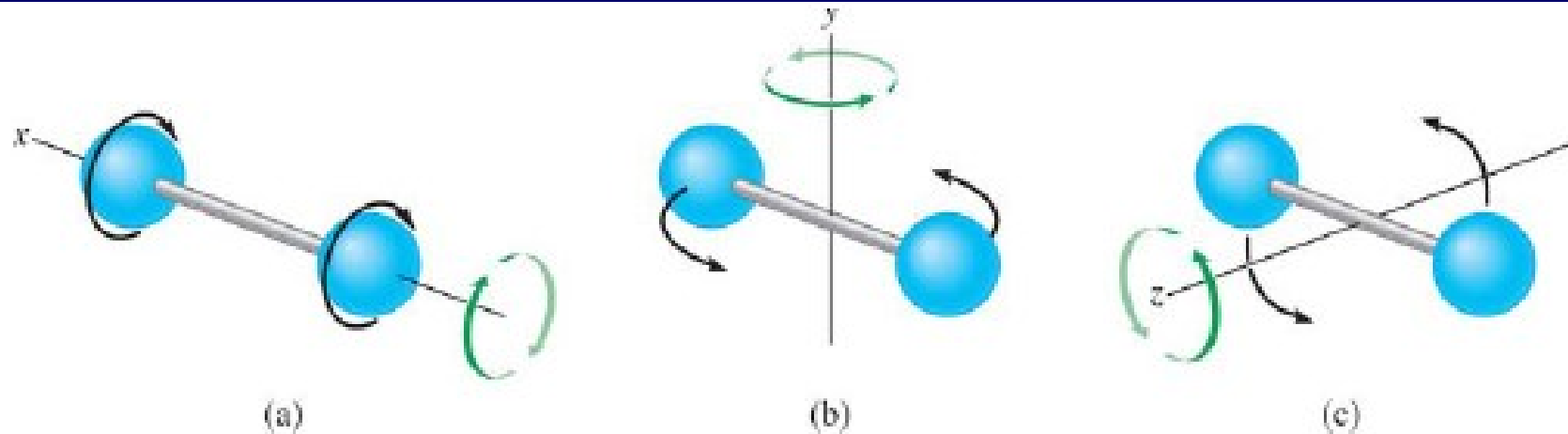
$$\overline{Bq^2} = \frac{1}{2} k_{\text{B}} T$$

Internal energy will be distributed equally among all possible degrees of freedom (equipartition of energy). Each degree of freedom contributes $\frac{1}{2} k_B T$ of energy per molecule and $\frac{1}{2} R$ to the molar specific heat at constant volume.

Diatomic molecule:

3 translational degrees of freedom

2 rotational degrees of freedom



Internal energy

$$U = N \times 5 \times \frac{1}{2} k_B T = \frac{5}{2} nRT$$

$$C_V = \frac{5}{2} nR$$

vibrational motion?

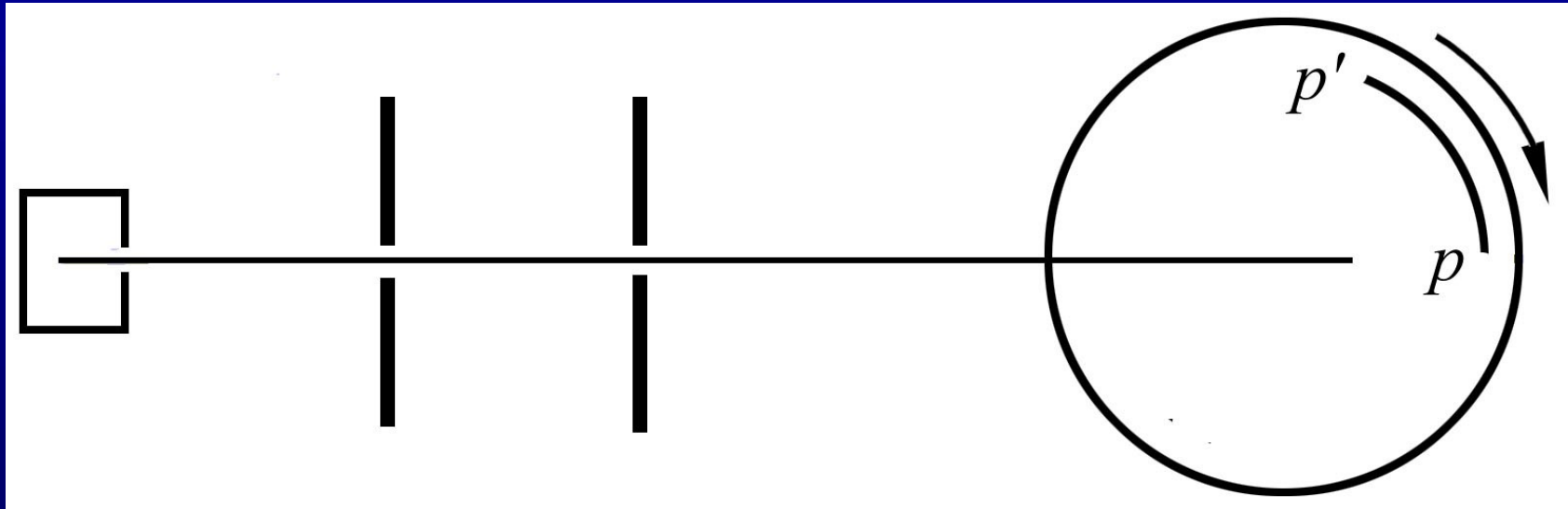
potential and kinetic components will both
contribute $\frac{1}{2} k_B T$

Diatomic gas	$C_{V,m}$ (J K⁻¹ mol⁻¹)	$C_{V,m} / R$
H ₂	20.18	2.427
CO	20.2	2.43
N ₂	19.9	2.39
Cl ₂	24.1	2.90
Br ₂	32.0	3.84

13.4 Effusion

泻流

‘Attempt of direct test of Maxwellian’



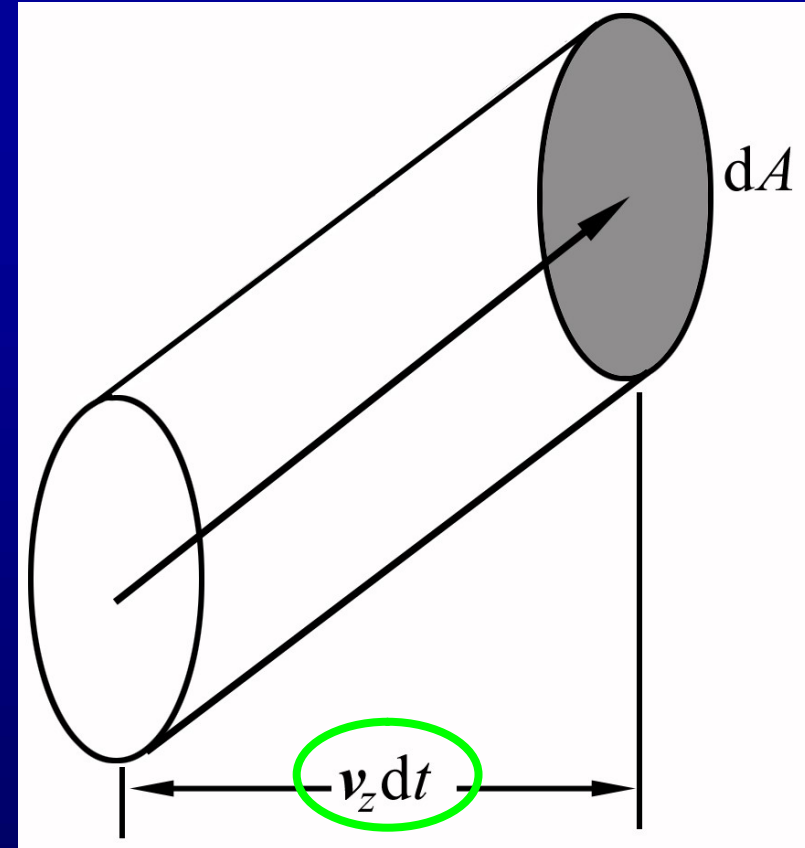
$$\Delta t = \frac{2R}{v}$$
$$\widehat{pp'} = R \cdot (\omega \Delta t) = \frac{2\omega R^2}{v} \longrightarrow v = \frac{2\omega R^2}{\widehat{pp'}}$$

$$dn \xrightarrow{\text{?}} d\Gamma$$

**the number of molecules
that hit the area dA in
interval dt is**

$$d\Gamma \underline{dA dt} = \iint_{v_x, v_y} v_z \underline{dt dA dn}$$

$$(v_z > 0)$$



$$d\Gamma(v_z) = n_0 v_z \left(\frac{m}{2\pi k_B T} \right)^{1/2} e^{-\frac{1}{2} m v_z^2 / k_B T} dv_z \quad (v_z > 0)$$

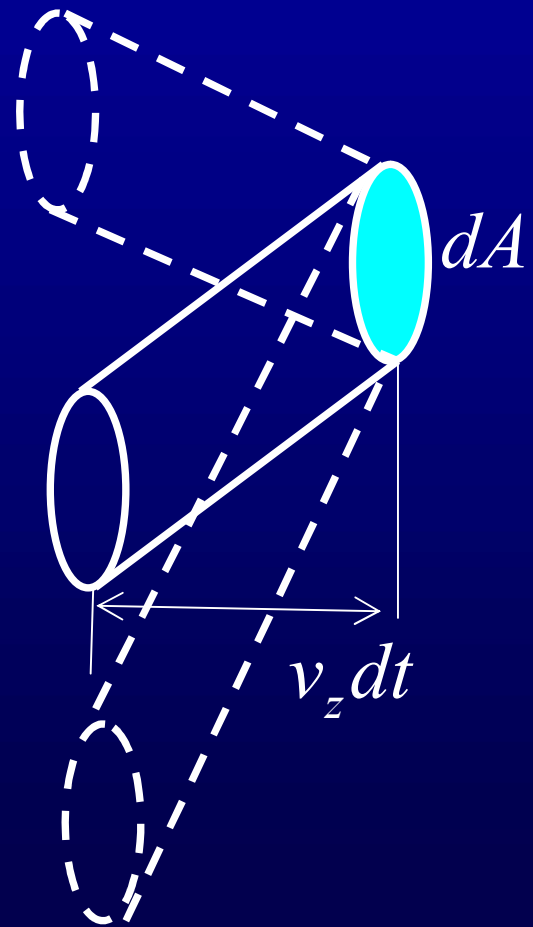
$$dn = (n_0 dA v_z dt) \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}} dv_x dv_y dv_z, \quad v_z > 0.$$

The number of molecules with z-component of velocity within $v_z \rightarrow v_z + dv_z$ that hit the area dA in time interval dt is

$$\int_{v_x, v_y} dn = (n_0 dA v_z dt)$$

$$\times \left(\frac{m}{2\pi k_B T} \right)^{3/2} \left(\int_{-\infty}^{+\infty} e^{-\frac{mv_x^2}{2k_B T}} dv_x \right) \left(\int_{-\infty}^{+\infty} e^{-\frac{mv_y^2}{2k_B T}} dv_y \right) e^{-\frac{mv_z^2}{2k_B T}} dv_z$$

$$= n_0 \left(\frac{m}{2\pi k_B T} \right)^{1/2} v_z e^{-\frac{mv_z^2}{2k_B T}} dv_z dA dt$$



$$\int_{v_x, v_y} dn = (n_0 dA v_z dt) \left(\frac{m}{2\pi k_B T} \right)^{3/2} \left(\int_{-\infty}^{+\infty} e^{-\frac{mv_x^2}{2k_B T}} dv_x \right) \left(\int_{-\infty}^{+\infty} e^{-\frac{mv_y^2}{2k_B T}} dv_y \right) e^{-\frac{mv_z^2}{2k_B T}} dv_z$$

$$= n_0 \left(\frac{m}{2\pi k_B T} \right)^{1/2} v_z e^{-\frac{mv_z^2}{2k_B T}} dv_z dA dt \equiv d\Gamma dA dt,$$

where $d\Gamma$ is the number of molecules with z-component of velocity within $v_z \rightarrow v_z + dv_z$ that hit on (or pass through) per unit area per unit time.

$$d\Gamma = n_0 \left(\frac{m}{2\pi k_B T} \right)^{1/2} e^{-\frac{mv_z^2}{2k_B T}} v_z dv_z = n_0 \frac{1}{\sqrt{\pi} v_m} e^{-\frac{v_z^2}{v_m^2}} v_z dv_z.$$

$$v_m = \sqrt{\frac{2k_B T}{m}}$$

Thus, the total number of molecules hit on (or pass through) per unit area per unit time (effusion flux 泻流密度) is

$$\begin{aligned}\Gamma &= n_0 \int_0^{\infty} \frac{1}{\sqrt{\pi} v_m} e^{-v_z^2/v_m^2} v_z dv_z \\ &= \frac{n_0}{2\sqrt{\pi}} v_m \\ &= \frac{n_0}{4} \bar{v}\end{aligned}$$

Quantities in equilibrium distribution

$$v_m = \sqrt{\frac{2k_B T}{m}}$$

$$\bar{v} = \sqrt{\frac{8k_B T}{\pi m}}$$

Speed distribution in the effusive beam

$$dn = (n_0 dA v_z dt) \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}} dv_x dv_y dv_z, \quad v_z > 0.$$

In spherical coordinates,

$$v_x^2 + v_y^2 + v_z^2 = v^2, \quad v_z = v \cos \theta,$$

$$dv_x dv_y dv_z \rightarrow v^2 \sin \theta dv d\theta d\varphi$$

$$dn = (n_0 dA v \cos \theta dt) \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-\frac{mv^2}{2k_B T}} v^2 \sin \theta dv d\theta d\varphi,$$

$$0 \leq \theta \leq \pi.$$

Using the Maxwell speed distribution

$$dn(v) = n_0 \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-\frac{1}{2} \frac{mv^2}{k_B T}} 4\pi v^2 dv \equiv f(v) dv$$

we have

$$dn = \frac{1}{4\pi} (v \cos \theta dA dt) f(v) dv \sin \theta d\theta d\varphi, \quad 0 \leq \theta \leq \pi.$$

$$dn = (n_0 dA v \cos \theta dt) \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-\frac{mv^2}{2k_B T}} v^2 \sin \theta dv d\theta d\varphi,$$
$$0 \leq \theta \leq \pi.$$

Thus

$$\begin{aligned}\int_{\theta, \varphi} dn &= \frac{1}{4\pi} dA dt v f(v) dv \int_0^{\pi/2} \cos \theta \sin \theta d\theta \int_0^{2\pi} d\varphi \\ &= \frac{1}{4\pi} dA dt v f(v) dv \cdot \frac{1}{2} \cdot 2\pi = \frac{1}{4} dA dt v f(v) dv \\ &\equiv d\Gamma dA dt,\end{aligned}$$

where $d\Gamma = \frac{1}{4} v f(v) dv$ is the number of molecules effused per unit area per unit time with speed $v \rightarrow v + dv$ in the effusive beam.

$$d\Gamma = \pi n_0 \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-\frac{mv^2}{2k_B T}} v^3 dv$$

The average speed of molecules in the effusive beam is

$$\begin{aligned} \bar{v}_\Gamma &= \frac{\int v d\Gamma}{\int d\Gamma} = \frac{\int e^{-v^2/v_m^2} v^4 dv}{\int e^{-v^2/v_m^2} v^3 dv} = v_m \frac{\Gamma\left(\frac{5}{2}\right)}{\Gamma\left(\frac{3}{2}\right)} \\ &= \frac{3}{2} \cdot \sqrt{\frac{2k_B T}{m}} = \frac{3}{2} \frac{\sqrt{\pi}}{2} \cdot \sqrt{\frac{8k_B T}{\pi m}} > \bar{v}. \end{aligned}$$

Assignment:13.10

13.5 Transport phenomena

Heat transfer : { conduction
convection (convective heat transfer)
thermal radiation

{ diffusion
viscosity

thermal conductivity

Conduction

$$\dot{Q} = J_Q A = -\kappa A \frac{dT}{dx}$$

$$J_Q = -\kappa \frac{\partial T}{\partial x}$$

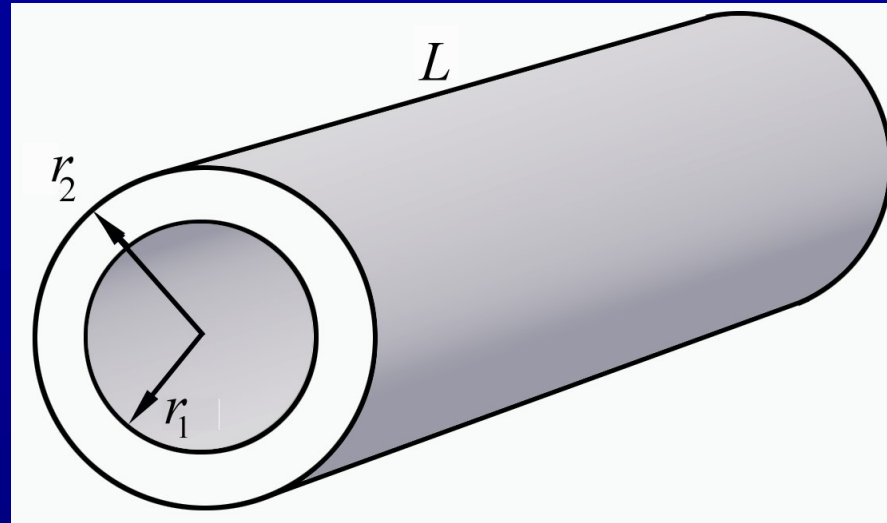
Fourier's law

Example

$$\dot{Q}_r = C = -\kappa 2\pi r L \frac{dT}{dr}$$

$$-\kappa \int_{T_1}^{T_2} dT = \dot{Q}_r \int_{r_1}^{r_2} \frac{dr}{2\pi r L}$$

$$\dot{Q}_r = \frac{2\pi\kappa(T_1 - T_2)L}{\ln \frac{r_2}{r_1}}$$



Convection

$$\dot{Q} = hA\Delta T$$

$h \sim$ density,
viscosity,
heat capacity,
heat conductivity,
speed of air,
laminar or turbulent flow, and
the curvature of the wall etc.

Quantitatively study of h is difficult but not quite necessary

Thermal radiation (from infrared to visible)

absorptivity α

polished surface of metal	0.03~0.08
lamp black	0.95
wet ice	0.97
blackbody	1

Stefan-Boltzmann Law:

$$J_B = \sigma T^4 \quad \sigma = 5.6703 \times 10^{-8} \text{ W}/(\text{m}^2 \cdot \text{K}^4)$$

Kirchhoff's law:

$$J_\alpha = \alpha J_B$$

On surface of the earth

$$J_{\text{ir}} = 1340 \text{ W}/\text{m}^2$$
$$= \alpha_{\odot} \sigma T_{\odot}^4 \frac{R_{\odot}^2}{D_{\text{ES}}^2}$$

$$J_r = \alpha_{\oplus} \sigma T_{\oplus}^4$$

$$\dot{Q} = A (\alpha_{\oplus} J_{\text{ir}} - J_r)$$

Diffusion

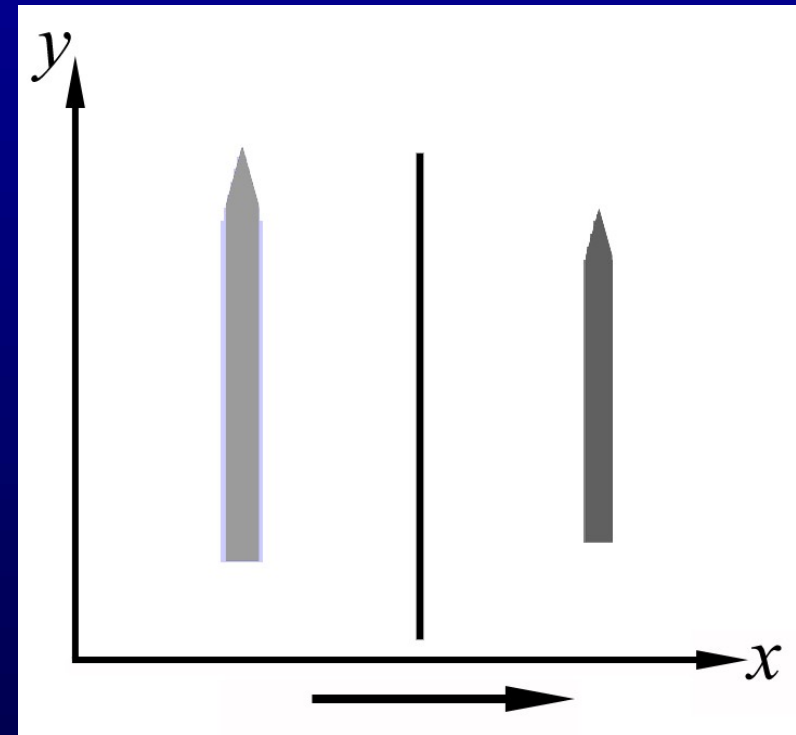
Diffusion coefficient

$$J_m = -D \frac{\partial \rho}{\partial x} \left(\frac{\partial p}{\partial x}, \frac{\partial T}{\partial x}, \Delta \mu \dots \right)$$

Fick's law

Viscosity

$$J_p = -\eta \frac{\partial v_y}{\partial x}$$



coefficient of viscosity

Conservation in transports

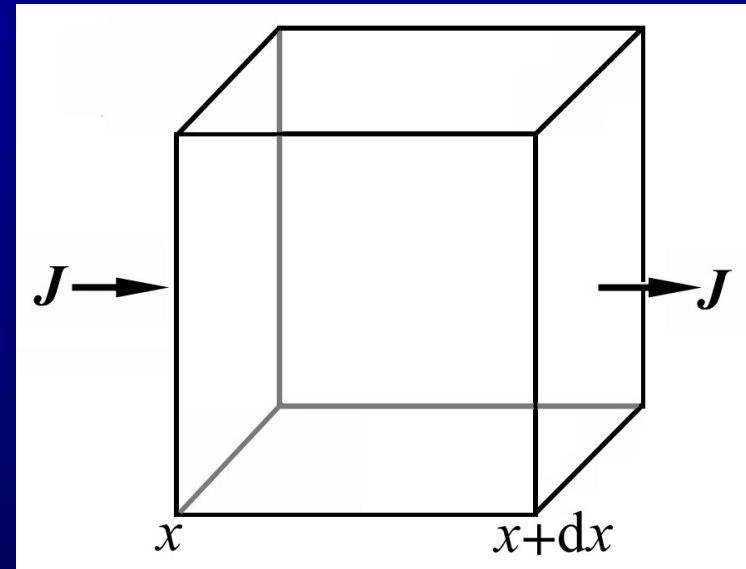
$$d\rho \, dA \, dx = J(x) \, dA \, dt - J(x + dx) \, dA \, dt$$

$$\approx -\frac{\partial J}{\partial x} \, dx \, dA \, dt$$

$$\frac{\partial \rho}{\partial t} = -\frac{\partial J}{\partial x}$$

$$\frac{\partial \rho}{\partial t} = D \frac{\partial^2 \rho}{\partial x^2}$$

$$J_m = -D \frac{\partial \rho}{\partial x}$$



$$\frac{\partial \rho}{\partial t} = D \frac{\partial^2 \rho}{\partial x^2} \qquad J_m = -D \frac{\partial \rho}{\partial x}$$

Similarly

$$\rho C_v \frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial x^2} \qquad J_Q = -\kappa \frac{\partial T}{\partial x}$$

$$\rho \frac{\partial v_y}{\partial t} = \eta \frac{\partial^2 v_y}{\partial x^2} \qquad J_p = -\eta \frac{\partial v_y}{\partial x}$$

source and sink

$$\frac{\partial \rho}{\partial t} = D \frac{\partial^2 \rho}{\partial x^2} + \boxed{C\rho}$$

For stationary state

$$\frac{\partial \rho}{\partial t} = 0, \quad \frac{\partial^2 \rho}{\partial x^2} + \frac{C}{D} \rho = 0$$

Neutrons in reactor

$$\frac{\partial^2 n}{\partial x^2} + B^2 n = 0$$

Standing wave solution ($n = 0$ at $x = 0, a$) is

$$n = N \sin(Bx) \quad \left(B = \frac{\pi}{a} \right)$$

Example 13.2 conduction equation with loss term

$$\frac{\partial T}{\partial t} = \frac{\kappa}{\rho C_V} \frac{\partial^2 T}{\partial x^2} - \alpha T$$

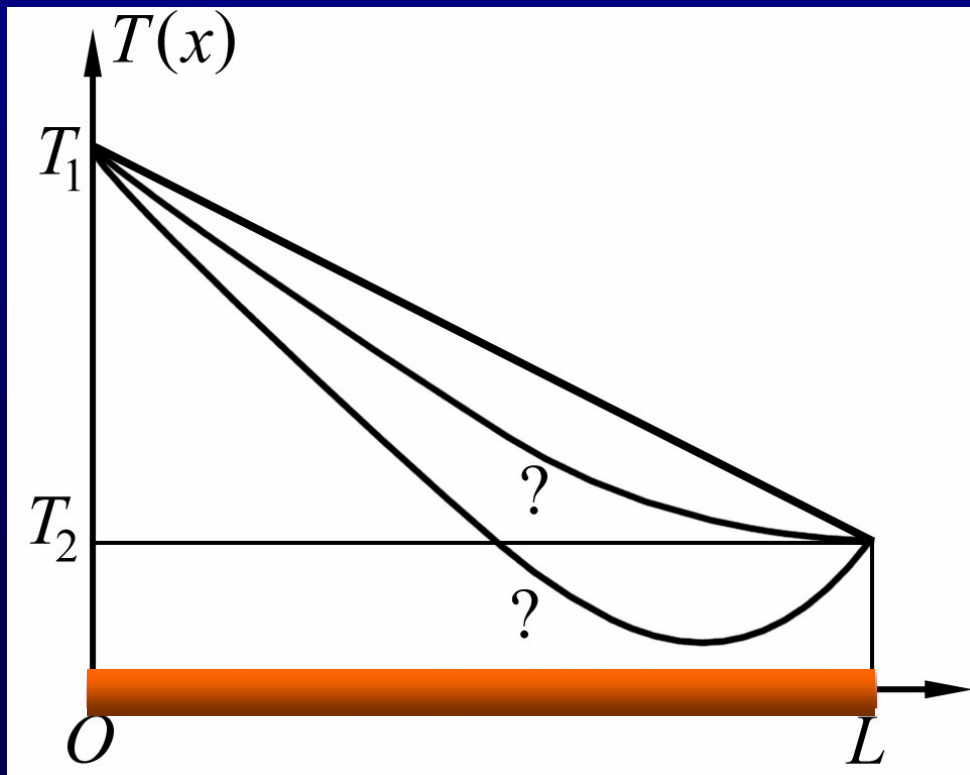
For stationary case

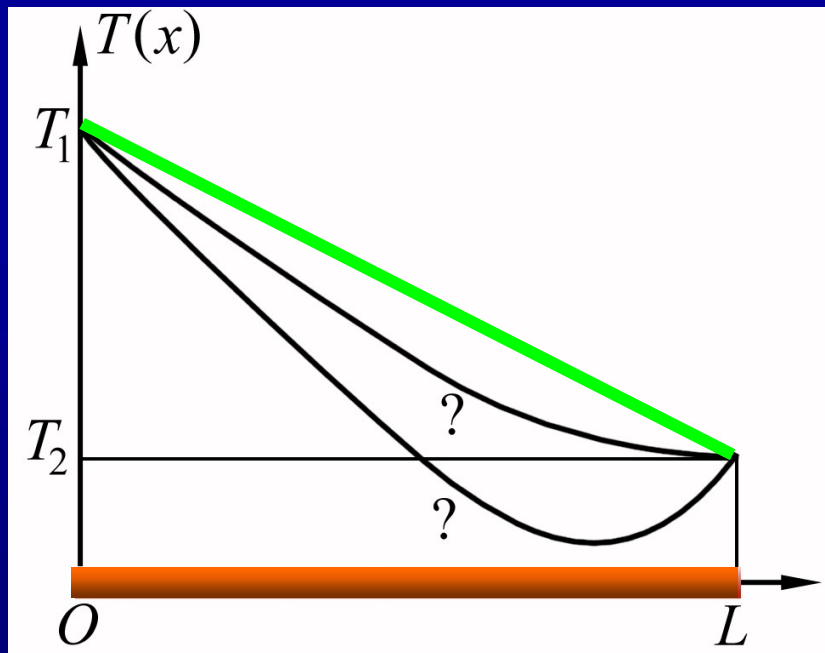
$$\frac{\partial^2 T}{\partial x^2} - \beta^2 T = 0$$

$$\beta^2 \equiv \frac{\rho C_V \alpha}{\kappa}$$

$$\alpha = 0$$

$$T = T_1 - (T_1 - T_2) \frac{x}{L}$$





$$\alpha \neq 0$$

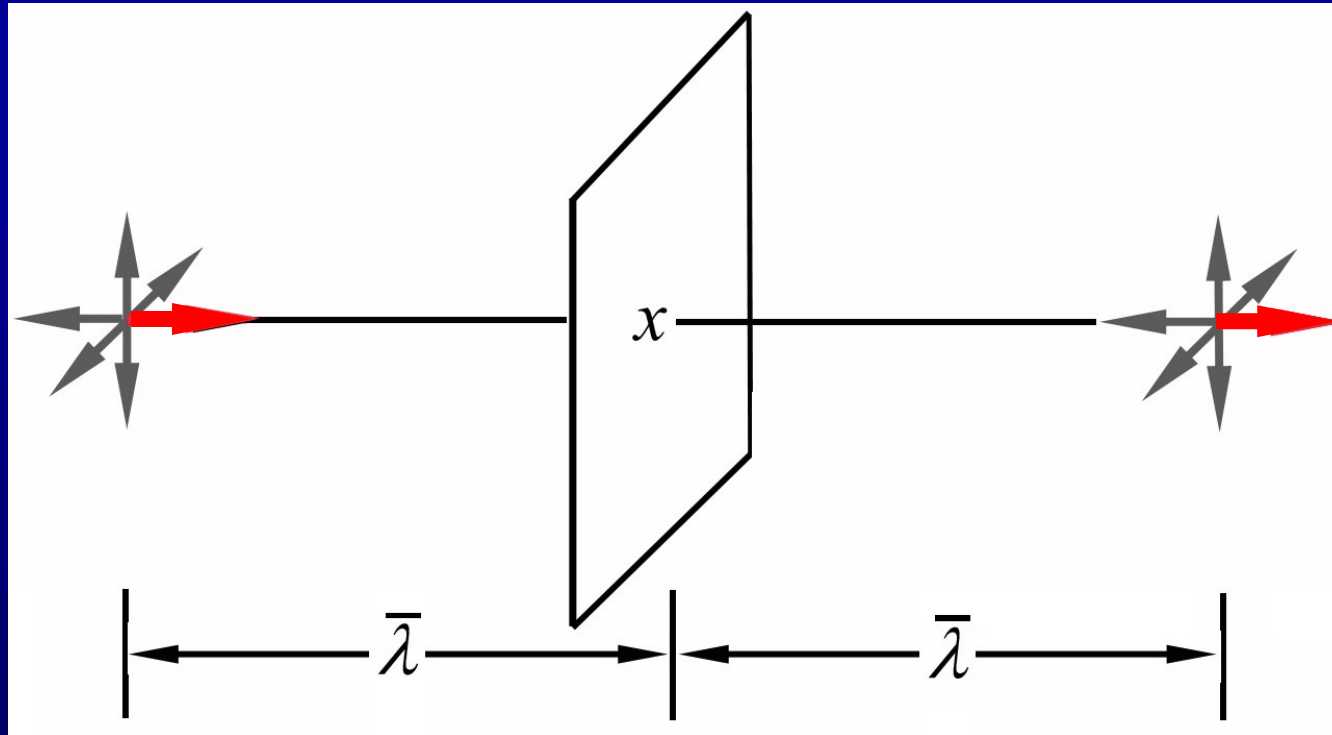
$$\frac{\partial^2 T}{\partial x^2} - \beta^2 T = 0$$

$$\beta^2 \equiv \frac{\rho C_V \alpha}{\kappa}$$

$$T = Ae^{\beta x} + Be^{-\beta x}$$

$$T = T_1 \frac{\sinh \beta (L - x)}{\sinh \beta L} + T_2 \frac{\sinh \beta x}{\sinh \beta L}$$

Microscopic theory of transport coefficients



$$\xi = m, \quad \frac{1}{2} m \overline{v^2} = \frac{3}{2} k_B T, \quad m v_y$$

$$\mathcal{E} = M, \quad Q, \quad p$$

$\bar{\lambda}$ -- mean free path 平均自由程

$$d\mathcal{E} = \frac{1}{6} dA \bar{v} dt \left[n(x - \bar{\lambda}) \xi(x - \bar{\lambda}) - n(x + \bar{\lambda}) \xi(x + \bar{\lambda}) \right]$$

For mass transport, $mn = \rho$

$$\begin{aligned} dM &= \frac{1}{6} dA \bar{v} dt \left[mn(x - \bar{\lambda}) - mn(x + \bar{\lambda}) \right] \\ &= \frac{1}{6} dA \bar{v} dt \left[-2\bar{\lambda} \frac{\partial \rho}{\partial x} \right] \end{aligned}$$

$$J_M \propto \frac{dM}{dA dt} = - \boxed{\frac{1}{3} \bar{v} \bar{\lambda}} \left(\frac{\partial \rho}{\partial x} \right)$$

$$D = \frac{1}{3} \bar{v} \bar{\lambda}$$

Similarly

$$\kappa = DC_V \left(n \frac{3}{2} k_B T = C_V T \right)$$

$$\eta = D\rho$$

Hydrogen under STP

$$m = 3.33 \times 10^{-27} \text{ kg}$$

$$n = p/k_B T = 2.68 \times 10^{25} / \text{m}^3$$

$$\bar{v} = \sqrt{\frac{8k_B T}{\pi m}} = 1.69 \times 10^3 \text{ m/s}$$

$$\bar{\lambda} = \frac{1}{\sqrt{2}n\sigma} \quad (5.3.17)$$

$$\sigma \sim 2r = 2 \times 1.12 \times 10^{-10} \text{ m}$$

$$\bar{\lambda} = 1.67 \times 10^{-7} \text{ m}$$

$$D = 9.42 \times 10^{-5} \text{ m}^2/\text{s},$$

$$\kappa = 8.71 \times 10^{-2} \text{ W}/(\text{m} \cdot \text{K})$$

Experimental results:

$$D = 1.28 \times 10^{-5} \text{ m}^2/\text{s},$$

$$\kappa = 1.68 \times 10^{-2} \text{ W}/(\text{m} \cdot \text{K})$$

Assignment: 13.12